

Telegraph

The network that turns intelligence into a graded commodity.

telegraphprotocol.com

The market for graded intelligence.

What it is

Like diamonds, intelligence needs a basis for grading. But intelligence keeps changing. Telegraph ranks models, APIs, datasets and tools within each Intent, making evolving performance comparable for buyers.

How the ranking works

Each Intent has a continuously updated ranking. Validators independently verify evaluation results and finalize ranking state through BFT consensus, with a 43-of-64 quorum at full genesis capacity.

Why it stays honest

Providers and Evaluators compete in an open market. Validators check committed evaluation evidence; fraud costs rewards and Validator eligibility.

Who does what

Providers compete to supply intelligence. Evaluators compete to improve how performance is measured. Validators verify and finalize rankings. Agents and applications create paid demand for intelligence.

How money moves

Buyers pay for intelligence in USDC. The protocol keeps 2%; the Provider share is aggregated for MACHINA purchases and Provider settlement on Ethereum.

The token and the proof

MACHINA: 21M cap, four-year halvings, zero pre-mine, zero team/VC allocation, proposed 25/50/25 split. Season I drew 1,400+ miners, 120K testnet transactions, and 45 apps. Mainnet Dec 26 / Jan 27.

Fixed supply. Graded intelligence.

- 21,000,000 Machina hard cap; 7,200 per 24-hour emission period at genesis; halving every 4 years; zero pre-mine, zero team / VC allocation.
- Proposed genesis split: Validators 25% / Evaluators 50% / Treasury 25%.
- Value linkage - fulfilled paid demand accumulates USDC for aggregated MACHINA purchases and Provider settlement. MACHINA is the economic unit of consumed intelligence, not a guaranteed return.
- Proposed immutable genesis: cap, halving, 25/50/25 split, zero pre-mine; other changes require the applicable governance process and timelock.

21M
hard cap, fixed

25 / 50 / 25
validator / evaluator / treasury

0%
miner emissions

Why operating a node is a major opportunity

Validators secure the changing grades of intelligence - the independently verified rankings that agents and applications use to choose intelligence for each task.

- Help secure a new graded-intelligence market as one of 64 genesis Validators.
- Verify the rankings that determine each Provider’s opportunity to serve paid demand.
- Earn Validator emissions and delegation commissions; separate Provider activity is optional.

Validator earnings, explained.

Validators earn emissions by performing verified work. If they attract delegated MACHINA, operators can also earn a commission on the resulting delegated rewards.

STREAM ONE / PROTOCOL

Verification emissions

- 25% of 7,200 = 1,800 MACHINA/day across the entire Validator pool.
- At 64 Validators, equal-share math is ~10,266 MACHINA/year per node; actual rewards are not equal by default.
- Payouts depend on completed assignments, internal audit allocation, eligibility and penalties.

STREAM TWO / OPTIONAL

Delegation commissions

- Delegated MACHINA can count toward a Validator's stake for active-set eligibility.
- Operators earn their configured commission from delegator-attributable rewards; the remainder goes to delegators.
- Commissions come from the same emission pool, not extra tokens. No Validator USDC fee share at genesis.

What a Validator can earn.

A fixed emission pool funds verification and consensus. The value of earned MACHINA depends on the token price, eligible completed work and the stake arrangement - not transaction volume alone.

GENESIS POOL / SCHEDULED EMISSIONS

25% of 7,200 MACHINA / day

ALL VALIDATORS, COMBINED

1,800 / day

ANNUALIZED NETWORK-WIDE

657,000 / year

Equal-share illustration (64): ~10,266 MACHINA/year per node. Real rewards vary with work and audit assignments.

HYPOTHETICAL MACHINA PRICES

Gross annual emission value per Validator, using ~10,266 MACHINA/year as an equal-share illustration.

MACHINA PRICE	ILLUSTRATIVE GROSS / YEAR
$0.10	$1,027
$0.50	$5,133
$1.00	$10,266
$2.00	$20,531

Illustrative only; not forecasts. Excludes costs and delegators.

WHY TELEGRAPH WINS

When intelligence gets cheaper, being right becomes more valuable.

Intelligence is abundant and keeps changing. Telegraph grades competing Providers per Intent, while Evaluators compete to improve the tests themselves. Validators independently finalize rankings that guide real paid demand. Without demand, a ranking is just a leaderboard; with demand, it becomes a market.

BUILD IT YOURSELF

$78M - $191M

to train one frontier model (GPT-4 to Gemini Ultra)

- AI / ML engineers at $160K-$300K each, per year
- $500K+ for a single 8-GPU server, plus power and ops
- 80%+ of AI projects fail to reach value (RAND)
- You still build verification and monitoring yourself

BUY GRADED INTELLIGENCE

$0.01

minimum request - intelligence ranked by Intent

- No training, no GPUs, no ML team to hire
- Rankings verified by independent Validators
- Higher-ranked Providers gain routing opportunity
- Buy intelligence per Intent, not model seats

MINER COST COMPARISON

How open supply becomes a graded intelligence market.

MACHINA connects paid demand to a market for continuously graded intelligence. Providers compete to serve each Intent; Evaluators compete to measure performance, and Validators finalize rankings that direct economic opportunity.

	BITCOIN	BITTENSOR ($TAO)	TELEGRAPH
What you do	Solve cryptographic puzzles.	Run models / provide intelligence in subnets.	Supply any intelligence behind an API (model, data feed, tool, etc.).
Hardware	ASICs ($4,000 - $10,000+ per machine).	High-end GPUs or dedicated cloud compute clusters.	Often none. Many miners just use an existing API + a YAML config file.
Ongoing cost	High electricity and cooling (~$150 - $600+ per month per machine; ~$32,000 to $59,000 in power per BTC mined).	Dynamic subnet slot registration fee (burned in TAO; typically $200 - $2,000+) + monthly GPU hosting ($300 - $2,000+/month).	Near zero. Main cost is whatever is already paid to run the existing API ($0 - $10/month serverless overhead).
Barrier	Very high.	High.	Extremely low.
Paid in	Block rewards + fees (emissions).	Subnet emissions (TAO).	Real demand only. Agents pay USDC; the Provider share buys MACHINA in aggregate and settles to the serving Provider. Zero Provider emissions.
Total cost to participate	$4,000 - $10,000+ upfront per ASIC + $150 - $600+/month recurring.	$500 - $2,000+ upfront + $300 - $2,000+/month recurring compute.	~$0 upfront + ~$0 - $100/month, depending on the kind of intelligence you provide.

Keys, provenance, consensus, and penalties

Key separation

Manager Key (cold) holds stake offline; Operator Key (hot) only signs. A breach can't touch principal; keys rotate on-chain.

Provenance (zkTLS)

zkTLS binds the Miner endpoint to the exact input and response; invalid or substituted evidence is rejected.

Consensus (BFT)

43 of 64 must sign to finalise (tolerates 21 bad/offline); stake-weighted median, commit-reveal, BLS-aggregated.

Credentials

Miner API keys encrypted across hardware enclaves; no single node holds the full key; auto-rotated on secure hardware.

Penalties

Liveness = ejection; fraud = forfeiture + permanent ban; incorrect execution = forfeiture; principal never slashed.

Abuse resistance

Economic Sybil cost on a fixed 21M supply; jittered TWAP settlement makes front-running unprofitable.

What it costs and how to deploy

PRE-MAINNET (CURRENT PHASE)

Running a node during testnet is extremely lightweight.

- Infrastructure - a standard server with 8-16 GB RAM; no GPUs or specialized hardware required.
- Cost - low hundreds of USD per month, plus a couple of testnet tokens.
- Setup - run a single script and bind a private key; no staking or capital at this stage.
- Staff - minimal oversight (uptime and key management).

Run a node immediately with almost zero financial commitment or operational complexity.

POST-MAINNET

Requirements shift to operating as a full validator.

- Capital - stake enough Machina to remain in the top 64 (market-determined, 21-day unbonding).
- Infrastructure - still lightweight, no GPUs; a secure enclave-capable server; submission gas reimbursed.
- Setup - stake via Manager Key (cold), bind an Operator Key (hot), deploy one config file, sync to NTP.
- Staff - minimal; uptime monitoring and basic key hygiene.

Compound further through delegated staking, running your own miners, and onboarding ecosystem agents.

Costs and break-even scenarios.

Illustrative annual economics for one eligible Validator. No Validator USDC transaction-fee share is assumed. Actual emissions depend on completed work, audit accounting and delegation.

GROSS TOKEN ILLUSTRATION	VALUATION SCENARIO	GROSS USD (HYPOTHETICAL)
~10,266 MACHINA / year	$1.00 / MACHINA	$10,266 / year

MONTHLY HOSTING	ANNUAL HOSTING	BREAK-EVEN TOKEN PRICE	NET AT $1 / MACHINA
$200 / month	$2,400 / year	$0.23	+$7,866
$400 / month	$4,800 / year	$0.47	+$5,466
$800 / month	$9,600 / year	$0.94	+$666
$1,500 / month	$18,000 / year	$1.75	-$7,734

READ THIS AS SENSITIVITY, NOT PROJECTED RETURNS.

Hosting costs are assumptions, not production benchmarks or supplier quotes. Figures exclude staffing, security, tax, stake opportunity costs, commission splits and execution volatility.

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